

Alexander Chin

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EDUCATION

University of California - San Diego

B.S. Mathematics-Computer Science

City College of San Francisco

A.S. Mathematics with Highest Honors

La Jolla, CA

Sep. 2025 – Jun. 2027

San Francisco, CA

Aug. 2024 – May. 2025

EXPERIENCE

Undergraduate Research Assistant

UC San Diego (URL Lab)

Jan. 2026 – Present

La Jolla, CA

- Conducted econometric regression analysis in STATA to evaluate the regional economic impacts of educational policy spillovers using data from IPUMS and the World Bank.
- Built scalable Python pipelines to geocode 400+ educational institutions and systematically tokenized and standardized 4,000 census records across various datasets for analysis

Research Intern

UC San Diego

Aug. 2025 – Sep. 2025

La Jolla, CA

- Investigated optimization dynamics in machine learning models by analyzing convergence behavior and training stability across varying hyperparameters
- Implemented and benchmarked multiple model configurations, achieving up to 92% classification accuracy

NASA Community College Aerospace Scholar (NCAS)

National Aeronautics and Space Administration

Jan. 2025 – Apr. 2025

Green Belt Maryland (Remote)

- Collaborated on multi-phase Mars mission design applying systems-level analysis to evaluate launch windows, landing constraints, propulsion limits, payload mass, risk mitigation, and budget trade-offs.
- Presented technical strategy and findings to NASA engineers and product managers.

Academic Tutor

City College of San Francisco

Jan. 2025 – May. 2025

San Francisco, CA

- Tutored 20+ students weekly in calculus and physics and held 10+ hours of office hours in collaboration with faculty.

PROJECTS

ClassGraph | *Python, Go, React, D3.js*

- Engineered Python and Go scraping pipelines to collect, parse, and transform 2,500+ courses into a structured SQL database with AST-backed representations
- Designed a TypeScript AST-based prerequisite resolution engine capable of parsing nested AND/OR logic trees and rendering dependency DAGs in D3.js.

TroupeRoute | *Python, OSMnx, OR-Tools, GeoPandas*

- Built a Python optimization system modeling multi-stop routing as a constrained Traveling Salesman Problem using road networks and travel-time matrices.
- Integrated OR-Tools to enforce time-window and overlap constraints, improving schedule feasibility and reducing total travel time.

webregICS | *Python, FastAPI, React*

- Developed a full-stack scheduling application with a FastAPI backend and React frontend to process structured academic schedule data and automate calendar generation
- Implemented backend recurrence and constraint logic to generate standards-compliant .ics files

TECHNICAL SKILLS

Languages: Python, Java, SQL, MATLAB

Frameworks & Tools: FastAPI, React, Git

Data & ML: Pandas, NumPy, Matplotlib, Plotly, Scikit-learn, PyTorch

Coursework: Data Structures & Algorithms, Probability, Stochastic Processes, Optimization, Linear Programming, Machine Learning